Inference for numerical data

Getting Started

Load packages

In this lab, we will explore and visualize the data using the **tidyverse** suite of packages, and perform statistical inference using **infer**. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let's load the packages.

```
library(tidyverse)
library(openintro)
library(infer)
```

The data

Every two years, the Centers for Disease Control and Prevention conduct the Youth Risk Behavior Surveillance System (YRBSS) survey, where it takes data from high schoolers (9th through 12th grade), to analyze health patterns. You will work with a selected group of variables from a random sample of observations during one of the years the YRBSS was conducted.

Load the `yrbss` data set into your workspace.

```
data('yrbss', package='openintro')
```

There are observations on 13 different variables, some categorical and some numerical. The meaning of each variable can be found by bringing up the help file:

```
?yrbss
```

1. What are the cases in this data set? How many cases are there in our sample?

13,583 cases

Remember that you can answer this question by viewing the data in the data viewer or by using the following command:

```
glimpse(yrbss)
```

```
## Rows: 13,583
## Columns: 13
## $ age                     <int> 14, 14, 15, 15, 15, 15, 15, 14, 15, 15, 15, 1~
## $ gender                  <chr> "female", "female", "female", "female", "fema~
## $ grade                   <chr> "9", "9", "9", "9", "9", "9", "9", "9", "9", ~
```

```
## $ hispanic                 <chr> "not", "not", "hispanic", "not", "not", "not"~
## $ race                     <chr> "Black or African American", "Black or Africa~
## $ height                   <dbl> NA, NA, 1.73, 1.60, 1.50, 1.57, 1.65, 1.88, 1~
## $ weight                   <dbl> NA, NA, 84.37, 55.79, 46.72, 67.13, 131.54, 7~
## $ helmet_12m               <chr> "never", "never", "never", "never", "did not ~
## $ text_while_driving_30d   <chr> "0", NA, "30", "0", "did not drive", "did not~
## $ physically_active_7d     <int> 4, 2, 7, 0, 2, 1, 4, 4, 5, 0, 0, 0, 4, 7, 7, ~
## $ hours_tv_per_school_day  <chr> "5+", "5+", "5+", "2", "3", "5+", "5+", "5+",~
## $ strength_training_7d     <int> 0, 0, 0, 0, 1, 0, 2, 0, 3, 0, 3, 0, 0, 7, 7, ~
## $ school_night_hours_sleep <chr> "8", "6", "<5", "6", "9", "8", "9", "6", "<5"~
```

Exploratory data analysis

You will first start with analyzing the weight of the participants in kilograms: `weight`.

Using visualization and summary statistics, describe the distribution of weights. The `summary` function can be useful.

```r
summary(yrbss$weight)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   29.94   56.25   64.41   67.91   76.20  180.99    1004
```

2. How many observations are we missing weights from?

1004 observations.

Next, consider the possible relationship between a high schooler's weight and their physical activity. Plotting the data is a useful first step because it helps us quickly visualize trends, identify strong associations, and develop research questions.

First, let's create a new variable `physical_3plus`, which will be coded as either "yes" if they are physically active for at least 3 days a week, and "no" if not.

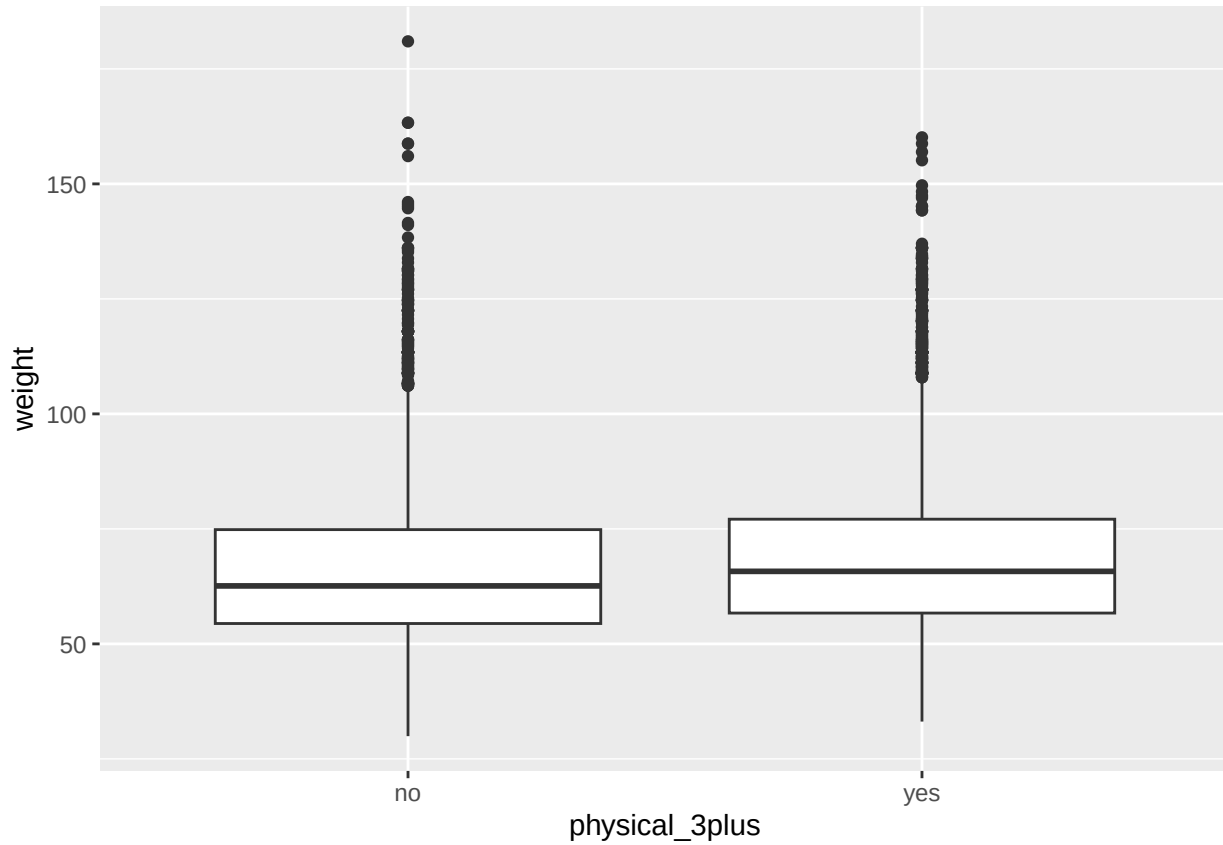
```r
yrbss <- yrbss %>%
  mutate(physical_3plus = ifelse(yrbss$physically_active_7d > 2, "yes", "no"))
```

3. Make a side-by-side boxplot of `physical_3plus` and `weight`. Is there a relationship between these two variables? What did you expect and why?

There does not appear to be a strong relationship between the two variables. I expected to see `physical_3plus = Yes` to have a lower weights, but the graph shows otherwise. The lower tail, the IQR box, the median, is all lower than yes.

```r
library(ggplot2)

yrbss |>
  drop_na(physical_3plus) |>
  ggplot(aes(x = physical_3plus, y = weight)) + geom_boxplot()
```



The box plots show how the medians of the two distributions compare, but we can also compare the means of the distributions using the following to first group the data by the `physical_3plus` variable, and then calculate the mean `weight` in these groups using the `mean` function while ignoring missing values by setting the `na.rm` argument to `TRUE`.

```
yrbss %>%
  group_by(physical_3plus) %>%
  summarise(mean_weight = mean(weight, na.rm = TRUE))
```

```
## # A tibble: 3 x 2
##   physical_3plus mean_weight
##   <chr>                <dbl>
## 1 no                    66.7
## 2 yes                   68.4
## 3 <NA>                  69.9
```

There is an observed difference, but is this difference statistically significant? In order to answer this question we will conduct a hypothesis test.

Inference

4. Are all conditions necessary for inference satisfied? Comment on each. You can compute the group sizes with the `summarize` command above by defining a new variable with the definition `n()`.

Yes, the conditions are satisfied. The YRBSS uses a multi-state cluster sampling for randomization. Observations for `physical_3plus` can only be yes/no making it independent, and both groups have n > 30, so that satisfies inference for `weight`.

5. Write the hypotheses for testing if the average weights are different for those who exercise at least times a week and those who don't.

h0 = The average weight for those who exercise at least 3 times a week is the same as those who don't. h1= The average weight for those who exercise 3 times a week is different than those who don't.

Next, we will introduce a new function, `hypothesize`, that falls into the `infer` workflow. You will use this method for conducting hypothesis tests.

But first, we need to initialize the test, which we will save as `obs_diff`.

```
obs_diff <- yrbss %>%
  drop_na(physical_3plus) %>%
  specify(weight ~ physical_3plus) %>%
  calculate(stat = "diff in means", order = c("yes", "no"))
```

Notice how you can use the functions `specify` and `calculate` again like you did for calculating confidence intervals. Here, though, the statistic you are searching for is the difference in means, with the order being `yes - no != 0`.

After you have initialized the test, you need to simulate the test on the null distribution, which we will save as `null`.

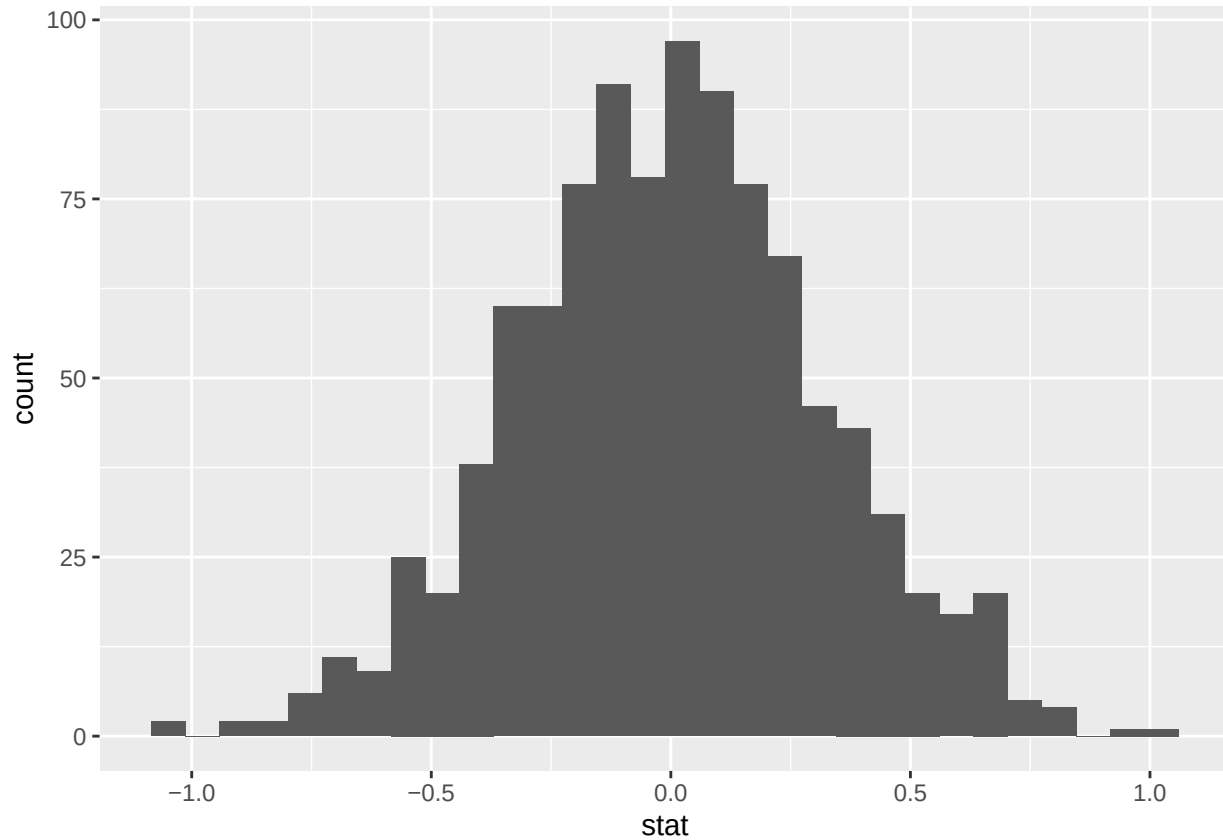
```
null_dist <- yrbss %>%
  drop_na(physical_3plus) %>%
  specify(weight ~ physical_3plus) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 1000, type = "permute") %>%
  calculate(stat = "diff in means", order = c("yes", "no"))
```

Here, `hypothesize` is used to set the null hypothesis as a test for independence. In one sample cases, the `null` argument can be set to "point" to test a hypothesis relative to a point estimate.

Also, note that the `type` argument within `generate` is set to `permute`, whichis the argument when generating a null distribution for a hypothesis test.

We can visualize this null distribution with the following code:

```
ggplot(data = null_dist, aes(x = stat)) +
  geom_histogram()
```



6. How many of these `null` permutations have a difference of at least `obs_stat`?

0

```
null_dist %>%
  summarize(answer = sum(stat >= obs_diff))
```

```
## # A tibble: 1 x 1
##   answer
##    <int>
## 1      0
```

Now that the test is initialized and the null distribution formed, you can calculate the p-value for your hypothesis test using the function `get_p_value`.

```
null_dist %>%
  get_p_value(obs_stat = obs_diff, direction = "two_sided")
```

```
## # A tibble: 1 x 1
##   p_value
##     <dbl>
## 1       0
```

This the standard workflow for performing hypothesis tests.

7. Construct and record a confidence interval for the difference between the weights of those who exercise at least three times a week and those who don't, and interpret this interval in context of the data.

lower_ci = 1.11 & upper_ci = 2.41, on average high schoolers in the U.S. who worked out 3 days or more weighed more on average than those who did not.

```
set.seed(123)

yrbss |>
  drop_na(physical_3plus, weight) |>
  specify(weight ~ physical_3plus) |>
  generate(reps = 1000, type = "bootstrap") |>
  calculate(stat = "diff in means", order = c("yes", "no")) |>
  get_confidence_interval(level = 0.95, type = "percentile")
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1     1.11     2.41
```

More Practice

8. Calculate a 95% confidence interval for the average height in meters (`height`) and interpret it in context.

lower_ci = 1.69 & upper_ci = 1.69, the average height for high schoolers in the U.S. is 1.69 meters.

```
yrbss |>
  drop_na(height) |>
  specify(response = height) |>
  generate(reps = 1000, type = "bootstrap") |>
  calculate(stat = "mean") |>
  get_confidence_interval(level = 0.95, type = "percentile")
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1     1.69     1.69
```

9. Calculate a new confidence interval for the same parameter at the 90% confidence level. Comment on the width of this interval versus the one obtained in the previous exercise.

It's the same. I guess the reasoning is that a 90% confidence interval is narrower, so if they are already the same at 95% this is the expected result.

```r
yrbss |>
  drop_na(height) |>
  specify(response = height) |>
  generate(reps = 1000, type = "bootstrap") |>
  calculate(stat = "mean") |>
  get_confidence_interval(level = 0.90, type = "percentile")
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1     1.69     1.69
```

10. Conduct a hypothesis test evaluating whether the average height is different for those who exercise at least three times a week and those who don't.

h0 = The average height of high schoolers in the U.S. who are physically active at least 3 days a week is the same from those who are not. h1 = The average height is different between the two groups. The p value is 0, so we can reject the null hypothesis and say that the average height of high schoolers who were physically active at least 3 days a week are different from those who are not.

```r
h_obs_diff <- yrbss |>
  drop_na(physical_3plus, height) |>
  specify(height ~ physical_3plus) |>
  calculate(stat = "diff in means", order = c("yes", "no"))

h_obs_diff
```

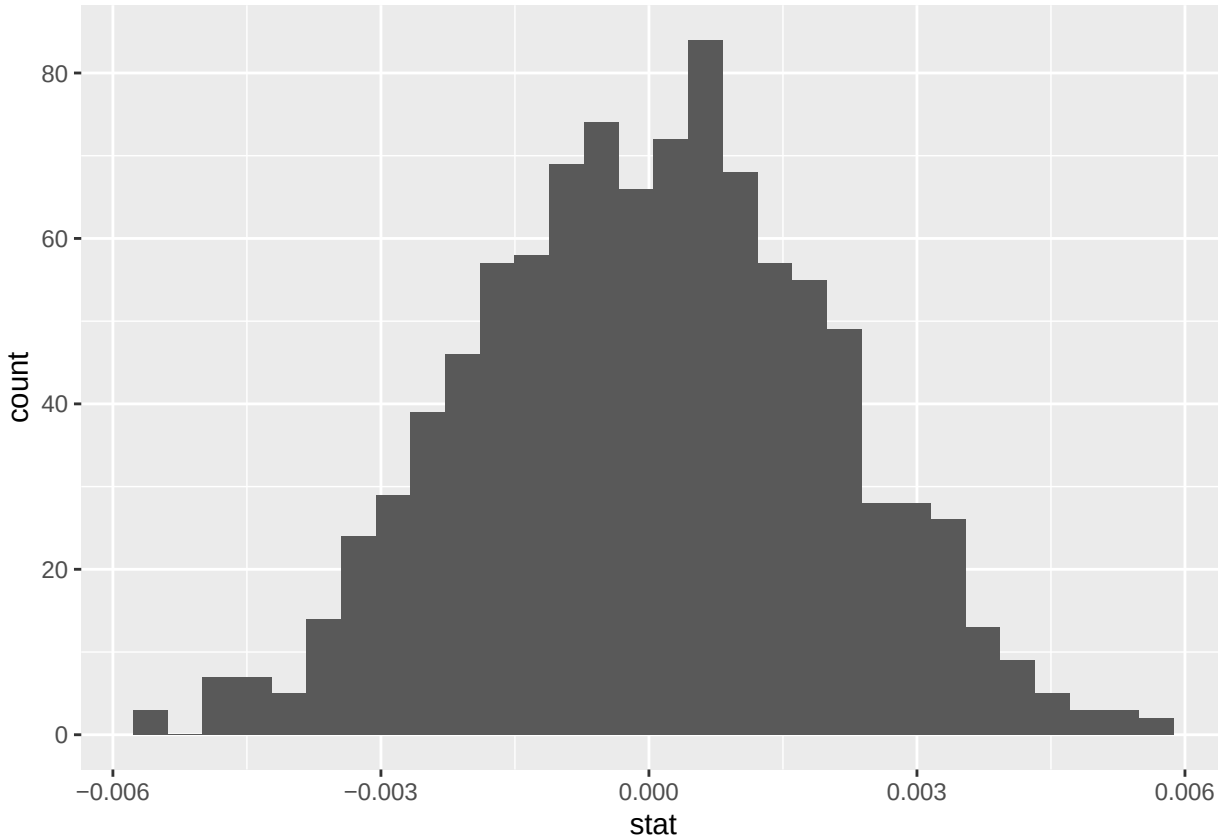
```
## Response: height (numeric)
## Explanatory: physical_3plus (factor)
## # A tibble: 1 x 1
##     stat
##    <dbl>
## 1 0.0376
```

```r
h_null_dist <- yrbss |>
  drop_na(physical_3plus, height) |>
  specify(height ~ physical_3plus) |>
  hypothesize(null = "independence") |>
  generate(reps = 1000, type = "permute") |>
  calculate(stat = "diff in means", order = c("yes", "no"))

h_null_dist |>
  get_p_value(obs_stat = h_obs_diff, direction = "two_sided")
```

```
## # A tibble: 1 x 1
##   p_value
##     <dbl>
## 1       0
```

```r
ggplot(data = h_null_dist, aes(x = stat)) +
  geom_histogram()
```



11. Now, a non-inference task: Determine the number of different options there are in the dataset for the `hours_tv_per_school_day` there are.

8 different options for hours_tv_per_school_day if you include NA.

```r
yrbss |>
  distinct(hours_tv_per_school_day) |>
  count()
```

```
## # A tibble: 1 x 1
##       n
##   <int>
## 1     8
```

12. Come up with a research question evaluating the relationship between height or weight and sleep. Formulate the question in a way that it can be answered using a hypothesis test and/or a confidence interval. Report the statistical results, and also provide an explanation in plain language. Be sure to check all assumptions, state your α level, and conclude in context.

Do U.S. high schoolers who sleep more than 8 hours a day have a different weight than those who sleep less than 8 hours a day? The p-value is 0 so we can reject the null hypothesis, high

schoolers who sleep more than 8 hours have a different weight than those students who don't sleep more than 8 hours.

```
yrbss <- yrbss |>
  mutate(sleep_8plus = ifelse(school_night_hours_sleep >= 8, "yes", "no"))
```

```
s_obs_diff <- yrbss |>
  drop_na(sleep_8plus, weight) |>
  specify(weight ~ sleep_8plus) |>
  calculate(stat = "diff in means", order = c("yes", "no"))

s_obs_diff
```

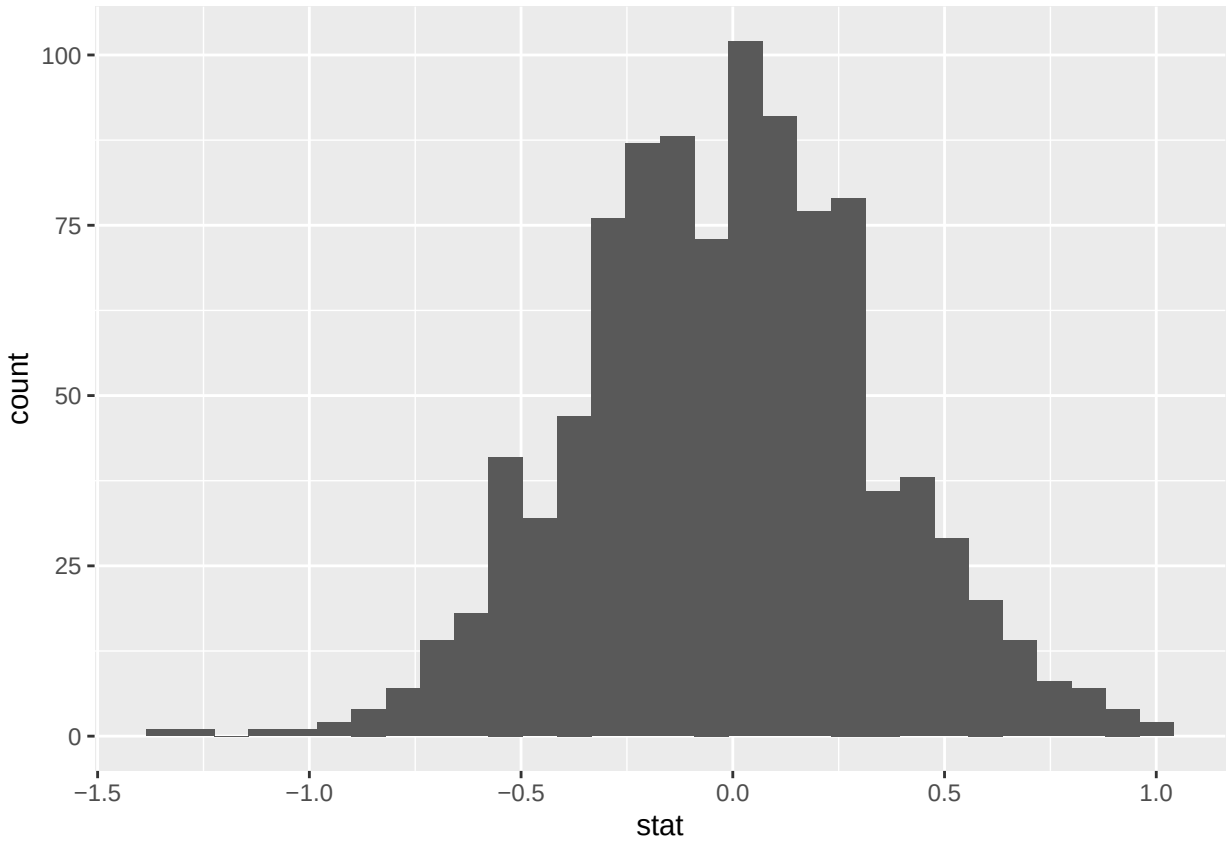
```
## Response: weight (numeric)
## Explanatory: sleep_8plus (factor)
## # A tibble: 1 x 1
##    stat
##   <dbl>
## 1 -1.18
```

```
s_null_dist <- yrbss |>
  drop_na(sleep_8plus, weight) |>
  specify(weight ~ sleep_8plus) |>
  hypothesize(null = "independence") |>
  generate(reps = 1000, type = "permute") |>
  calculate(stat = "diff in means", order = c("yes", "no"))

s_null_dist |>
  get_p_value(obs_stat = s_obs_diff, direction = "two_sided")
```

```
## # A tibble: 1 x 1
##   p_value
##     <dbl>
## 1   0.004
```

```
ggplot(data = s_null_dist, aes(x = stat)) +
  geom_histogram()
```


100
75
50
25
0
count
−1.5
−1.0
−0.5
0.0
0.5
1.0
stat